Trolling
Interim Project Critique

HDForge

22.04.2024

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Trolling

HDForge

Fan Yang, Jiexu Xie, Shideh Shasti, XiangLong Lin, Xiaoran Lyu, Xinzhe Hou
Design Computing Studio 3 – Proposal
22.04.2024

1 Background Research

With more research, we decided to focus on "trolling." Trolling is the act of intentionally posting provocative or offensive statements online. This issue needs to be addressed urgently as it is becoming more prevalent and trolling spreads from user to user, triggering an overall deterioration of the community if the trolling behaviour carries over from previous discussions (Cheng et al., 2017).

Trolling is difficult to define; it is a complex behaviour that has only been studied in a few academic studies and does not have a unified definition. Generally speaking, trolling can be categorised into three types, when trolling first appeared on the Internet, it was only used for the purpose of making fun of each other in the online community, a way to welcome new members, and this type of trolling was called classic trolling. However, over time, the purpose of trolling has gradually changed, and instead of being a fun and friendly behaviour, it has become a malicious version known as anonymous trolling (Bishop, 2014). Classic trolling rarely exists anymore, and most of the trolling on the internet today are anonymous trolling, but in addition to these two types, there is also a specific type of politically motivated trolling. "Trolls" with political agendas actively encourage violent attacks against opposition leaders and journalists, especially during critical political periods (Binark, 2015). Another noteworthy type of "troll" is an individual hired by an organisation to manipulate public opinion, often referred to simply as an "opinion manipulation troll" (Mihaylov et al.)

When a "trolling" incident occurs, three important groups of users are involved: perpetrators, victims and bystanders. As for the perpetrators, they are the ones who post malicious statements to the victims and they are the most immature and unsympathetic people whose purpose is to attract attention or just for fun. People with a lot of screen time are more likely to be perpetrators (Andreassen et al., 2013), in the context of gaming, trolls are usually younger, mostly male and spend more time playing games every day (Thacker & Griffiths, 2012). One study found that trolling has a similar pattern of relationship with the Dark Tetrad of personality, which is positively correlated with sadism, psychopathy, and Machiavellianism, which is the strongest correlation between sadism and trolling behaviours, and that trolling can be described as an Internet manifestation of everyday sadism (Buckels, 2014). Bullies commonly have lower academic performance, more health issues, and offline challenges, while also displaying narcissistic traits, so many of the perpetrators are also victims (Nicol. 2012).

A trolling bystander is an individual who witnesses trolling, they are the ones who are influenced by trolling and the ones who can influence trolling. Bystanders are an easy group to overlook in trolling situations, although there is a great deal of research that shows this group has an impact on the behaviour of cyberbullying, but there is still a gap in research focus on trolling‘s bystanders. However, since trolling is a type of cyberbullying, bystander behaviour, influence, etc. should be similar, and we can still refer to studies on bystanders of cyberviolence. Studies show that between 59 and 70 percent of college students have witnessed

cyberbullying on social media (Gahagan et al., 2016), but most people don't react positively; they usually ignore the information (Doane et al., 2020), this type of bystanders is called an outsider and is one of the three bystander types, the remaining two being the defender and the reinforcer (Zhao et al.,2023). Reinforcer may exacerbate or contribute to acts of cyberbullying; conversely, defender is who take positive action to stop or mitigate cyberbullying. Komac and Cagiltay (2019) did research on the effect of witnessing trolling behaviour on the mood of bystanders and found no correlation between the two.

Regarding victims, research has shown that more severe manifestations of taunting, such as repeated harassment and identity-based insults, can subject victims to similar outcomes as cyberbullying, including anxiety and depression (Gray et al., 2017). Trolling can attack anyone, but the people among the age of middle school are susceptible to trolling (Nicol. 2012), that's why we wanted to consider them as a target audience for designing digital solutions. They are in an important phase of developing identity and autonomy, specifically in the fifth stage of Erikson's 8 stages of psychosocial development, which is identity vs confusion (Blos, 1967; Dunkel & Harbke, 2017). Recognizing this vulnerability, we have designed a questionnaire tailored specifically to collect responses from them. The purpose of this questionnaire was to understand adolescents' experiences and impacts of trolling on social media in order to develop appropriate solutions.

2 Analysis of the results of the questionnaire

2.1 Close-ended Questions

Based on the analysis of the questionnaire results of close-ended questions, we can see that teenagers are heavy users of social media, with the majority spending more than 2 hours a day on social media. 75% of the participants indicated that they had experienced trolling, which indicates that trolling is a prevalent problem on social media. Regarding forms of trolling, the top three included provocation, sarcasm and speech interference. In addition, half of the population reported experiencing trolling sometimes, with the majority of behaviours coming from strangers. The top three responses to troll were ignore it, block the troll, and confront the troll directly. Besides, more than half of the participants felt that trolling affected their activities and interactions on social media, and even affected their offline life and mental health. Finally, when asked which platform they experienced the most trolling on, many participants identified Twitter as the main platform. Although many felt that social media platforms have taken effective measures to combat the problem, they felt that there are still areas that need to be strengthened.

2.1 Open-ended Questions

With regard to descriptions of specific examples, some participants described direct attacks and humiliations they had experienced on social media, such as attacks on their bodies, opinions or personal lives. Others described discrimination against specific groups or provocations on sensitive topics. Some responses also mentioned the creation of fake accounts and false and harmful content. When they experience trolling, the main emotions are anger, depression, disappointment, insecurity, sadness, confusion, anxiety, frustration, speechlessness, and so on. In addition, the negative effects of trolling on the lives of victims include impaired mood, decreased sleep quality, threats to personal safety, decreased trust, and impact on social interactions.

3 Impact on Society

The impact of trolling on society is pervasive and deeply concerning, with far-reaching consequences across various domains. Research indicates that trolling not only perpetuates a toxic online culture but also has tangible effects on individuals' mental health and well-being. For instance, studies have shown that exposure to trolling behavior can lead to increased levels of stress, anxiety, and depression among victims, particularly adolescents who are still developing their emotional regulation skills (Coyne et al., 2019). Furthermore, the anonymity afforded by the internet emboldens trolls to engage in increasingly aggressive and harmful behavior, exacerbating the problem (Buckels et al., 2014).

The societal impact of trolling extends beyond individual harm to encompass broader implications for digital citizenship and public discourse. Trolling undermines the credibility of online platforms as reliable sources of information and fosters a climate of distrust and scepticism among users (Massanari, 2017). This erosion of trust not only diminishes the quality of online interactions but also undermines democratic principles by impeding meaningful civic engagement and deliberative discourse (Marwick & Lewis, 2017).

To address these challenges, stakeholders must adopt a multi-faceted approach that addresses the root causes of trolling while also mitigating its effects on individuals and society. For users, education and empowerment are key strategies. Providing adolescents with resources and guidance on digital literacy, online safety, and emotional resilience can help mitigate the impact of trolling and empower them to navigate online spaces more effectively (Nocentini et al., 2019). Bystander intervention programs, modelled after successful initiatives in offline settings, can equip users with the tools to recognize and address trolling behavior within their communities.
Platforms and policymakers also play a crucial role in combating trolling and fostering healthier online environments. Implementing robust content moderation policies backed by transparent enforcement mechanisms can help curb trolling behavior and promote civility. Additionally, technological solutions such as AI-driven moderation tools can augment human efforts by identifying and removing harmful content at scale. Strengthening regulations around online anonymity and accountability can further deter trolling by increasing the risk of detection and consequences for perpetrators

By addressing the societal roots of trolling and implementing targeted interventions at multiple levels, stakeholders can work towards creating safer and more inclusive online spaces where individuals can express themselves freely without fear of harassment or harm.

4 Further Strengthening the Legal Framework

In the proposed amendments to the Telecommunications Act 1997 ((Department of Infrastructure, Transport, Regional Development, Communications and the Arts, 2017), the definition of "cyber-trolling" is proposed to be clarified to include but not limited to provocative, ironic, and discriminatory words and actions. This idea is based on our team's analysis of the results obtained from questionnaire surveys, which is consistent with most of the literature and research results on the Internet, indicating that a series of behaviours frequently mentioned and reported in these feedbacks are considered to cause significant problems. The amendment aims to repair the loopholes in existing legal provisions in the face of all forms of trolling to comprehensively improve legal protection for users. Correspondingly, penalties that are commensurate with the severity of the crime should also be set, especially for serious

acts that cause psychological trauma, in order to protect judicial fairness and justice on behalf of victims and accurately curb potential criminals.

In the proposal to modify the Criminal Code Act 1914 (Department of Infrastructure, Transport, Regional Development, Communications and the Arts, 2021a), it is advocated to expand the scope of its legal definition to include both indirect and direct forms of digital harassment, in order to combat the complexity of online behaviour that is not yet covered by the current legal definition. This plan will prevent criminals from maliciously exploiting legal loopholes to better protect citizens from the harm of emerging digital trolling forms. In addition, strengthening legal cooperation at the international level is also worth keeping up with when dealing with foreign-related trolling cases, because the Internet era has brought long-distance and convenient communication, and trolling often involves international people and actors. The existing legal framework should strengthen cross-border accountability.

To further strengthen the Enhancing Online Safety Act 2015 (Department of Infrastructure, Transport, Regional Development, Communications and the Arts, 2021b), it is necessary for the eSafety Commissioner to be empowered to freeze or delete harmful trolling content more quickly and proactively launch follow-up investigations. However, the current authority is not sufficient to respond quickly and effectively to serious online trolling. The strengthening of the Commissioner's rights is more likely to stop the spread of harmful content in the first place, which will greatly reduce the potential harm to victims. In addition, social media platforms such as Twitter, Instagram, and Facebook, which have been repeatedly mentioned in the questionnaire survey results, should also be required to update and maintain advanced detection algorithms in an effort to proactively identify and reduce online trolling misconduct. This requirement increases the responsibility of the platform and actively protects user security.

The above 3 legal framework reforms and expansions are stricter and more comprehensive user protections against trolling in the face of new challenges in the digital age, aiming to protect users from harassment, create a safer online environment, and drive the dual development of law and technology.

5 Ethical Consideration

To ensure that the final solution is effective, fair, and respects the rights and well-being of all users, it was evident that several ethical considerations should be taken into account prior to selecting a design direction from our conceptual model to address trolling on Twitter. Consequently, through evaluating the pros and cons for each of the 4 potential solutions mapped in our conceptual model at this stage of the project, key ethical considerations from a UI and UX perspective were underlined:

5.1 Identity Verification:

This is a concept that has both potential benefits and drawbacks. In terms of benefits, firstly, user identity verification can lead to increased accountability in its anonymity reduction as users are less likely to engage in abusive or harassing behaviour when their real identity is attached to their actions. Next, the credibility of the platform can be drastically enhanced as verified accounts may be perceived as more trustworthy, which can ultimately encourage healthier user interactions. Finally, this has the potential for safety enhancement on the platform, where identity verification can provide an added layer of security for users often targeted by trolls.

Regardless of the pros, as with all initial, abstract solution concepts, there are cons that require consideration before further action is taken. In this instance, requiring users to verify their identities raises privacy concerns. Many users value the anonymity provided by online platforms and may be reluctant to share personal information. In fact, it is highly possible that some may argue that reducing anonymity of online platforms takes away the freedom of online culture. This then brings up the issues of user exclusion for those who cannot or do not want to share personal information, as well as creating potential for abuse, where determined trolls may still find ways to harass others through potentially using stolen identities or creating fake accounts. This raises further technical challenges with the risk of false flagging or identity theft if the system is not properly designed and maintained. Finally, at this larger scale that a platform like Twitter operates, there are many global considerations that need addressing as identity verification norms vary across different cultures and countries where what may be acceptable in one context could be considered invasive or unacceptable in another.

5.2 Detection System

The concept of improving detection systems seems very promising in its journey to fulfil our design project's final desired outcome of reducing the prevalence of trolling on Twitter. Firstly, the idea of real-time intervention where an advanced detection system can identify and flag potentially abusive or harassing content automatically as it happens, allows for quicker intervention by platform moderators. Moreover, with swift identification and removal of trolling behaviour, these systems can reduce the harm caused and create a safer environment for users. Additionally, while the automated system can flag potentially abusive content, human moderators can review these cases for context and make more nuanced decisions which creates a balanced approach to content moderation. Through this, the machine learning algorithms can continuously learn and adapt, allowing the trolling detection system to improve over time and become more effective at identifying abuse.

As for areas of consideration, firstly, automated detection systems may incorrectly flag non-abusive content as trolling, leading to censorship of legitimate speech. Similarly, trolls are adept at finding ways to bypass such detection systems by using evasive techniques such as coded language. As a result, keeping up with evolving trolling strategies can be challenging for detection algorithms. Finally, developing and maintaining an effective trolling detection system can be very resource intensive in its requirement for ongoing monitoring and refinement.

5.3 User Education

Educating users (both potential victims and perpetrators) on Twitter can be a valuable approach to reduce trolling and foster a healthier online community. For instance, educating users about the impacts of trolling and how to recognise and respond to it can empower individuals to protect themselves and others from online abuse. Next, by raising awareness about the consequences of trolling and the importance of respectful online interactions, education efforts can help change the behaviour of individuals who may engage in abusive behaviour unknowingly or impulsively. Then, through this acknowledgement of the human impact of online harassment, empathy can be fostered among users and encourage them to be more considerate of others before engaging in abusive behaviour. Ultimately, unlike reactive measures such as content moderation or blocking, an education initiative has the potential to create long-term cultural shifts in online behaviour which sustainably reduces trolling and harassment over time.

Despite these potential benefits, the concept harbours certain drawbacks worth mentioning. For example, there is the concern of resistance or scepticism where some users deeming it unnecessary or intrusive may be resistant to educational messages about online behaviour, while others may be sceptical of the effectiveness of education in preventing trolling. Furthermore, education initiatives may face challenges in reaching users from diverse cultural or linguistic backgrounds. Trolling behaviour can also be influenced by ranging factors such as anonymity, group dynamics, and power imbalances. As such, educating users about the complexity of trolling dynamics is challenging and may require ongoing support. Finally, while education can play a valuable role in preventing trolling, it is certainly unable to be sufficient as a stand alone solution but may be helpful as a complementary measure to an (2, 4) advanced moderation system or (2) technical intervention to address trolling effectively.

5.4 Improved Reporting System

Improving the reporting system on Twitter is a crucial aspect of combating online trolling and incorporates benefits from all 3 of the other ideas covered above. For example, similar to (3) User Education, an improved reporting system can again, empower users to take action against trolling by providing them with accessible and effective tools to report the abusive behaviour. Next, much like (2) Detection System, a well-designed reporting system enables platform moderators to efficiently review and address reports of trolling which leads to quicker removal of abusive content and enforcement of community guidelines. This can help mitigate the harm caused to victims and deter perpetrators from engaging in further abuse. Ensuring a long-term solution at reducing trolling, reporting data can also provide valuable insights into the prevalence and nature of trolling on the platform, helping inform (1) policy decisions, (2, 4) algorithmic improvements, and (3) educational initiatives aimed at preventing future instances of abuse. Finally, a transparent and responsive reporting system fosters trust and confidence (1) among users, encouraging greater and more positive engagement with and within the platform.

Nevertheless, areas for ethical consideration exist and must be addressed if this concept is to be developed further. Firstly, large online platforms like Twitter receive an overwhelming volume of reports each day which makes it undoubtedly challenging for moderators to review and prioritise them effectively. As such, it may be very resource intensive to scale moderation efforts accordingly. Additionally, there is the risk of moderators tasked with reviewing reports bringing their own biases and subjective interpretations to the process, leading to inconsistent or unfair outcomes. As a result, great investments need to be made in training and oversight efforts to ensure fair and impartial moderation decisions. Next, improving reporting systems may inadvertently encourage perpetrators to abuse the system by filing false reports against innocent users or engaging in mass reporting campaigns to silence dissenting voices. In which case, it is crucial for the platform to implement safeguards to detect and prevent such misuse. Finally, the complexity of trolling behaviour makes it challenging to accurately identify and address through reporting systems alone. With this in mind, to effectively address trolling, Twitter must complement reporting systems with other measures, such as (2) automated detection algorithms and (3) user education initiatives.

Conclusively, in order to effectively address trolling on Twitter, a multifaceted approach is needed, combining all, if not most of the 4 potential ideas of Identity Verification, Automated Detection Systems, User Education, and Improved Reporting Systems. Each concept offers unique benefits, from increasing accountability through identity verification to empowering users to report abusive behaviour. However, ethical considerations such as privacy concerns,

technical challenges, and cultural barriers must be navigated carefully to ensure the final proposed solution can effectively reduce trolling on Twitter for target users and work towards creating a safer and more inclusive online environment with minimal drawbacks.

6 Design Criteria

1. Quick Loading: Make sure the website loads fast so people don't get frustrated waiting for it to appear.
2. Mobile-Friendly: Ensure the website looks good and works well on phones and tablets too.
3. Easy to Use: Make it simple for everyone to navigate around the site and find what they need without any trouble.
4. Accessible to Everyone: Ensure that people with disabilities can use the website easily.
5. Easy to Find on Google: Make sure the website shows up when people search for related topics.
6. Keep People Interested: Make sure people spend time on the website and don't leave right away.
7. Encourage People to Take Action: Make it easy for people to do things like sign up for newsletters or make purchases.
8. Interesting and Helpful Content: Provide content that people find useful and easy to understand.
9. Keep People's Information Safe: Make sure any personal information people share on the website is protected from hackers.
10. Make Sure the Website Works Properly: Ensure the website is always up and running smoothly without any glitches.
11. Follow the Rules: Make sure the website follows all the laws and regulations related to things like privacy and security.

7 Design Opportunities

In the part of Design Opportunities, we will delve into specific strategies and interventions that aimed at reducing trolling among teenagers and therefore enhancing their online safety and experience.Compared to cyberbullying, trolling among teenagers has some special characteristics：Trolling may be more covert, using humour or sarcasm to make it difficult for the victim to detect, but it may still cause serious psychological harm. Teenagers may interact more frequently and intensively on social networks, making them more susceptible to trolling. Teenagers may have a lower level of awareness of trolling behaviour and may be more susceptible to mental health impacts when faced with trolling behaviour.

Therefore, design opportunities need to pay special attention to balancing the concealment and harm of trolling behaviour among teenagers: being able to identify and deal with trolling in a timely manner without excessively restricting freedom of speech; providing mental health support and emotional support to help them cope with psychological distress; Strengthen education and awareness raising: teach teenagers correct online behaviours and coping strategies to increase their awareness and vigilance against trolling; and meet their special needs in social support and interactive experience, encourage them to share and talk with others, and reduce the consequences of trolling reduce social pressure, protect the physical and mental health of teenagers, and improve their online sense of security.

Before designing specific solutions, research and evaluation are firstly needed to have a better understanding of the development of trolling therefore to implement interventions that are regularly evaluated to ensure the implementation and timeliness of strategies.

Research and evaluation specifically includes the following steps:
Design of in depth research and evaluation methods: Create an organised way of researching cyberbullying to figure out what causes it and how it works. You could use things like surveys, talking to people, group chats, looking at what's being said online, and more to really get what's going on.
Data collection and analysis tools: Make tools for collecting data and ways to analyse it for the research. This means making online surveys, data collection stuff, text checking software... to get and look at lots of different kinds of data and find helpful stuff in it.
Experimental and intervention design: Make up experiments and checks for different interviews to see if they work okay. This might mean setting up test groups, making plans to fix things, and gathering info from tests... to see what works best. Strategy optimization and improvement: Look at what we find from research and checks, then make plans to make things better. This could mean changing how we fix things, making sure we use stuff right, updating rules... to keep making things better and lasting longer.
Knowledge sharing and dissemination: Come up with a good way to share what we find from research and check with people who need to know and everyone else. This might mean having meetings, writing up what we find, suggesting things to do, using social media and websites to tell people... to make sure everyone knows about cyberbullying and how important it is to pay attention to.
Therefore comes into the following design opportunities:
AI technology powered Moderation Tools:
Problem/Needs: There is a large amount of virtual bullying in current social media platforms and online communities, but there is a lack of effective ways to identify and deal with these problems timely.
Improvement points: It is necessary to design an AI technology which can automatically identify and filter virtual trolling content to reduce the negative impact on teenagers and improve the security and user experience of the platform.
Key factors: accuracy and efficiency. AI technology needs to accurately identify virtual trolling and be able to process them quickly with real time requirements in order to protect users from virtual trolling. In addition, platforms need to ensure that the operation of AI tools do not misjudge legitimate content to avoid negative impacts on users.
User friendly Reporting Mechanisms:
Problem/Needs: Current reporting mechanisms may be complex or difficult to operate, resulting in users being reluctant to report virtual bullying.
Improvement points: Design a more user-friendly reporting process,which includes simplifying operation steps and providing clear guidance and feedback mechanism to encourage users to actively participate in reporting virtual bullying.
Key factors: simplicity and transparency. The reporting process should be simple and easy for users to complete. In addition, platforms should provide a clear feedback mechanism to let users know that their reports have been received and processed, thereby enhancing users' trust in and willingness to use the reporting mechanism.
Educational Initiatives:
Problem/Need: Teenagers lack sufficient awareness and coping skills of online security and are prone to become victims of trolling.

Improvement points: Design lively, interesting and interactive educational resources, and improve teenagers' online safety awareness through various forms of educational activities and courses, and enhance their ability to identify and respond to trolling.
Key factors: interactivity and relevance. Educational resources and activities should be engaging and interactive, sparking interest and engagement among young peoples. At the same time, educational content should provide practical information and coping strategies based on the characteristics of trolling and the needs of teenagers to enhance their online security awareness and capability.
Psychological Support Services:
Problem/Needs: Teenagers affected by trolling lack psychological support and effective coping strategies and are prone to emotional distress and mental health problems.
Improvement points: Design psychological support services specifically for adolescents, including providing personalised counselling and treatment plans, establish support groups and online communities, and carry out mental health education activities to provide emotional support and help to victims.
Key factors: personalization and continuity. Psychological support services should provide personalised help and treatment options based on the individual needs and circumstances of the victim. In addition, services should be ongoing and able to support victims over the long term, helping them to gradually recover their mental health and effectively cope with the effects of virtual bullying.
Community Engagement Efforts:
Problem/Needs: All parts of society have insufficient attention and participation in the issue of trolling, and there is lack of effective cooperation and coordination.
Improvement points: Design various forms of community participation projects, including holding community activities, launching volunteer projects and establishing partnerships, to promote participation and cooperation from all walks of life and jointly deal with the problem of trolling.
Key factors: participation and sharing. Community participation projects should attract broad participation from all walks of life and be able to build community consensus and an atmosphere of joint efforts. In addition, projects should encourage the sharing of knowledge and experiences and promote interaction and collaboration among community members to achieve a comprehensive response to the problem of trolling.
Legal Frameworks and Research:
Problem/Needs: Current legal frameworks and policy measures may be insufficient to effectively address the rapidly evolving problem of trolling and lack scientific research and data support.
Improvement points: Design a framework to support cooperation between research institutions and legal institutions, and provide data support and policy recommendations to formulate more effective legal frameworks and policy measures to protect the rights and safety of young people.
Key factors: scientificity and adaptability. Legal frameworks and policy measures should be based on scientific research and data support to ensure their effectiveness and sustainability. In addition, the law should be able to promptly respond to emerging forms and challenges of trolling and protect the rights and safety of young people.
"Note: The report of design opportunities was partly written in Chinese and translated into English using google translate, with grammar adjustments made using Grammarly.
The conceptual diagram created using Miro."

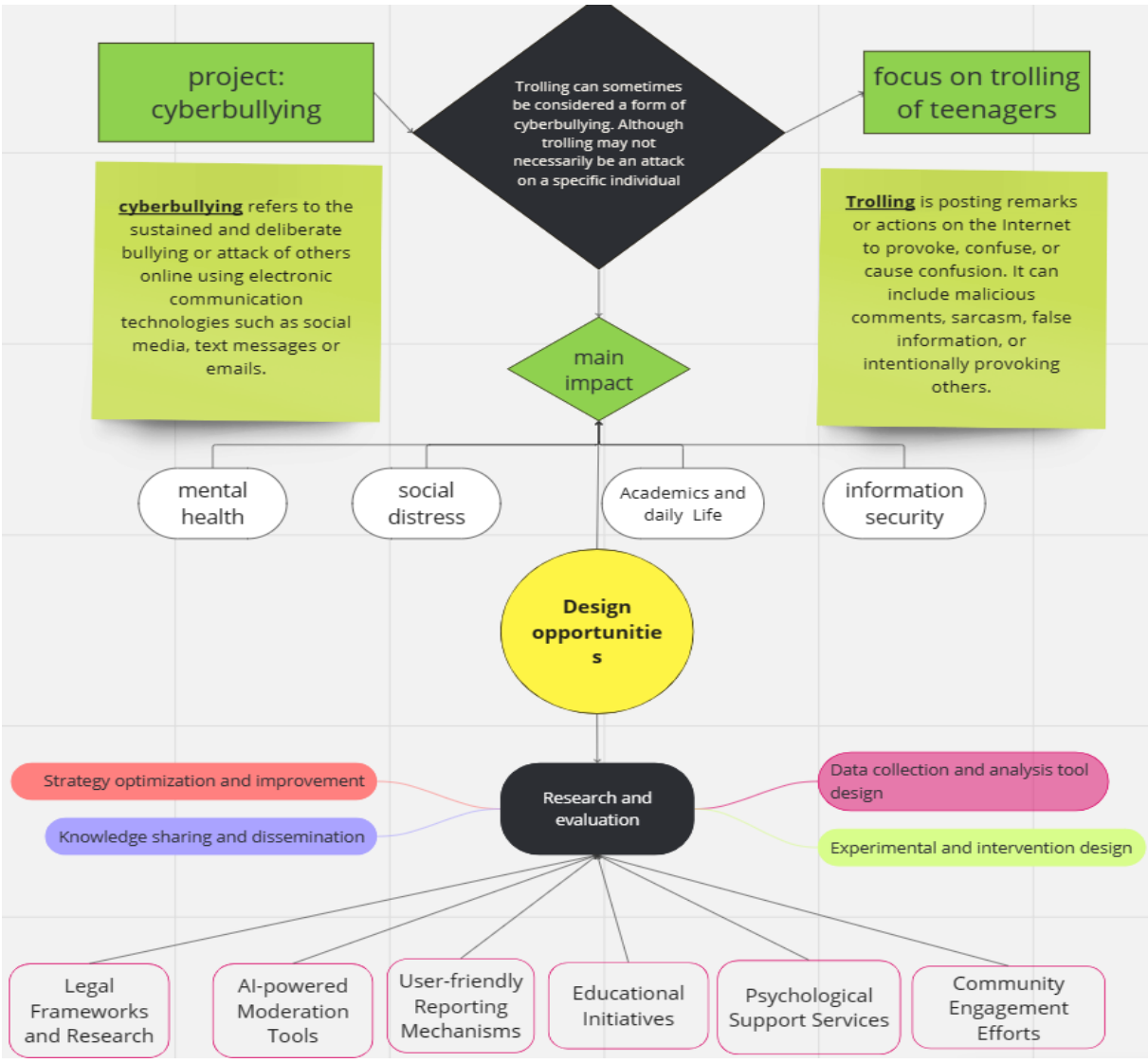

Figure 1 shows a diagram (made using Miro) of identified design opportunities for this project.

8 Prototype & Workflow

To aid us in gathering data in support of our developing design direction, as well as to answer any UI and UX questions identified, the following prototype process flowchart was created based on our conducted user research.

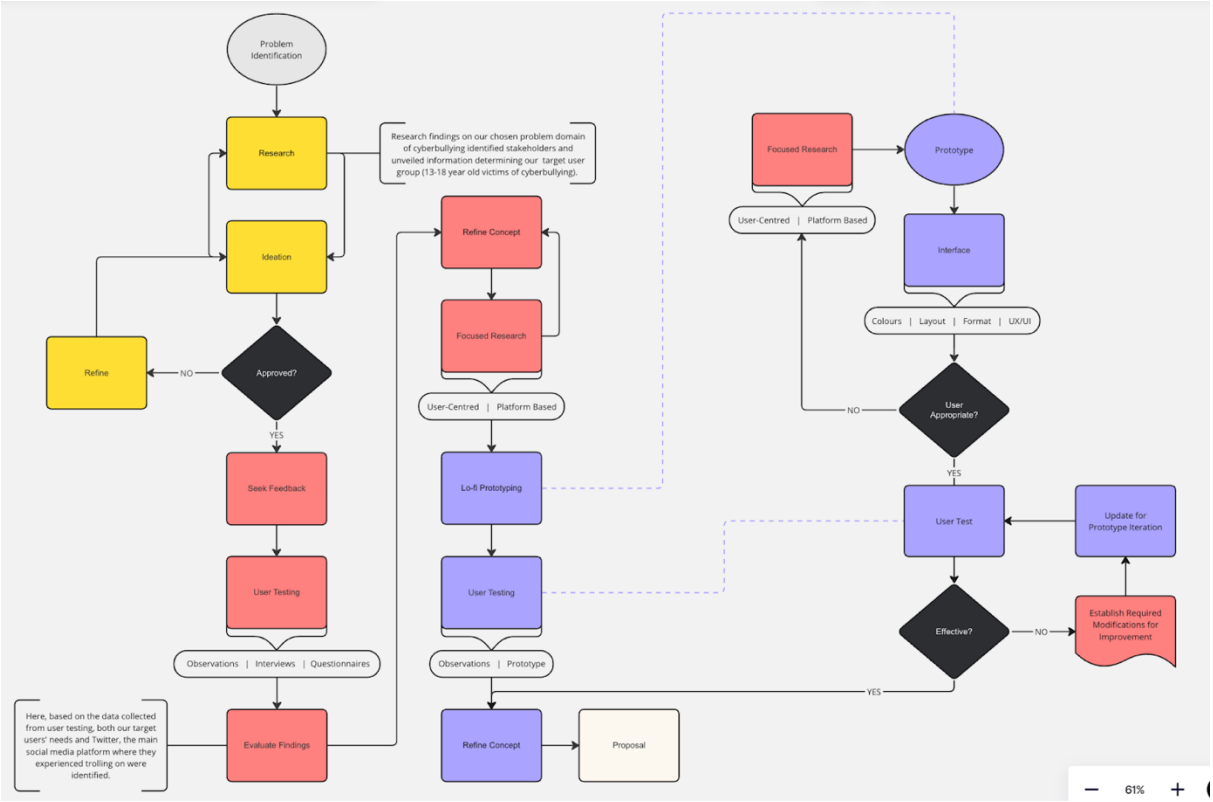

Figure 2 shows a flowchart (made using Miro) depicting the prototype process for this project.

The flowchart above depicts our workflow and intentions of producing a low fidelity prototype to elicit further, focused feedback from stakeholders and explore the true feasibility of particular solutions in our conceptual model (*refer to Figure 2*). The prototyping process will enable us to improve our knowledge about both the problem and solution space, ensuring we are well adept to propose a final solution which will be effective in tackling the project's desired outcome of reducing trolling on Twitter for target users.

9 Feedback

After presenting our Problem Identification we have received a lot of valuable feedback from our peers and tutors. Their insights provided a new perspective on our projects and highlighted some of the areas that need to be improved.

One crucial feedback we got from the tutor was the need for a specific problem space. In our original plan, the problem space was too general, we tried to focus on the prevention of cyber harassment, but this wouldn't have worked because with such a broad range our target audience would have been impossible to identify and for each type of cyber harassment there exists a different solution so we can't have just a single solution that tries to solve everything, this feedback has to lead us to refine our background and user research to reconsider a new specific problem space.

Another valuable critique was related to our target audiences. Peers Pointed out that our target audiences are too vague because we have two focuses we tried to design for the victims and offender, those two are very different, and to identify which ones to focus on we have observed the trolling behaviours on social media platforms and utilised surveys to gather insights from our target audiences. One problem we faced was that gathering information from

the victims was relatively simple, but when we tried to get a perspective from the offender's side was difficult, so we decided to just design for the victims.

Additionally, feedback highlighted that we are focusing too much on coming up with a solution rather than just walking through the design progress step by step. To address this, we have taken a step back and tried to use prototyping skills to explore the entire solution space as opposed to what we did before just creating a product for our conceptual model. We have listened to the tutor's suggestions and planned design activities such as personas, storyboards, Wizard of Oz and so on. Those will help us learn more about the problem space to explore and discover more potential design opportunities.

10 Team Reflection

10.1 Direction of the Project

Our team has been working hard to combat cyberbullying, especially around the problem space of trolling. We have done a lot of background and user research to get a deeper understanding of the needs of our users and design opportunities. There are also many challenges our team faced, for example, the legal aspects of the problem since we are trying to prevent trolling we need to also understand the ethical issues for the users, one of our possible solutions is using a filter method from AI to determine whether a comment from a user is just a normal comment or does it include hateful trolling messages, if it does include do we hide this comment from every else or allow it to be shown due the idea of freedom of speech we need to carefully balance around those, So there are still many problems and challenges to be solved we plan to explore the problem space further and identify more aspects and do additional investigations to collab with our stakeholders.

10.2 Team Process

Our team has excellent communication and collaboration skills each of our team members has utilised their own skill set to tackle the task. It is important to have different skills among each member, so we have to view things from a different perspective. We have established good communication channels and weekly meetings, however, we know that there is still improvement in terms of coordination. We might need to improve on taking weekly meeting logs to keep track of what has been discussed and provide more feedback to improve the efficacy and quality of the projects.

11 Project Plan

As we progress through Weeks 9-12 of our project, we encounter numerous opportunities to refine our design process. We have changed quite a bit since our last presentation. We were focusing on the final solution of the project, but after we listened to the feedback from the tutor and peers we changed our direction and tried to take a step back to get a better understanding of the problem space and user requirements. Our next step will be a list of activities to prepare ourselves for prototyping. We’re going to design personas, so we can have a visual view of our target audiences instead of an abstract range of people that we highlighted, we also will create the storyboards for our users, so we would understand the needs of our users to get into their perspective. Finally, we conduct the Wizard of Oz. This method allows our team to test designs that are powered by complex technologies at low cost as we can't implement those functions due to limited knowledge. It will provide us with early insights into the

project's utility and usability. After we have done those activities above. Through iterative prototyping and usability testing, we can test multiple iterations of the product and rapidly receive feedback on each idea. Then we will have an activity to collaborate with stakeholders, such as educational institutions and technology companies to guide our decision-making process.

12 References

Andreassen, C. S., Griffiths, M. D., Gjertsen, S. R., Krossbakken, E., Kvam, S., & Pallesen, S. (2013). The relationships between behavioral addictions and the five-factor model of personality. *Journal of Behavioral Addictions*, *2*(2), 90–99. https://doi.org/10.1556/jba.2.2013.003

Binark, M., Karataş, Ş., Çomu, T., & Koca, E. (2015). Türkiye'de Twitter'da trol kültürü. *Toplum Bilim, 135*, 124-157.

Bishop, J. (2014). "Representations of ' trolls ' in mass media communication: a review of media-texts and moral panics relating to ' internet trolling '", *Int. J. Web Based Communities, 10*(1), 7-24.

Blos, P. (1967). The second individuation process of adolescence. *Psychoanalytic Study of the Child, 22*, 162-186.

Buckels, E. E., Trapnell, P. D., & Paulhus, D. L. (2014). Trolls just want to have fun. *Personality and Individual Differences*, *67*, 97–102. https://doi.org/10.1016/j.paid.2014.01.016

Cheng, J., Bernstein, M., Danescu-Niculescu-Mizil, C., & Leskovec, J. (2017). Anyone Can Become a Troll: Causes of Trolling Behavior in Online Discussions. *arXiv.org*. https://doi.org/10.48550/arxiv.1702.01119

Coyne, S. M., Stockdale, L., Linder, J. R., Nelson, D. A., & Collier, K. M. (2019). Do psychopathic traits buffer the effects of online trolling behavior? Cyberpsychology, Behavior, and Social Networking, 22(2), 116-120.

Department of Infrastructure, Transport, Regional Development, Communications and the Arts. (2017). Enhancing Online Safety Act 2015 (Superseded version C2017C00187 (C03) from 23 June 2017 to 11 April 2018). Retrieved from https://www.legislation.gov.au/C2015A00024/2017-06-23/text

Department of Infrastructure, Transport, Regional Development, Communications and the Arts. (2021). Crimes Act 1914 (Superseded version C2021C00127 (C136) from 17 February 2021 to 31 August 2021). Retrieved from https://www.legislation.gov.au/C1914A00012/2021-02-17/text

Department of Infrastructure, Transport, Regional Development, Communications and the Arts. (2021). Telecommunications Act 1997 (Superseded version C2021C00134 (C99) from 17 February 2021 to 16 June 2021). Retrieved from https://www.legislation.gov.au/C2004A05145/2021-02-17/text

Doane, A. N., Ehlke, S., & Kelley, M. L. (2020). Bystanders Against Cyberbullying: a Video Program for College Students. *International Journal of Bullying Prevention*, *2*(1), 41–52. https://doi.org/10.1007/s42380-019-00051-5

Dunkel, C. S., & Harbke, C. (2017). A Review of Measures of Erikson's Stages of Psychosocial Development: Evidence for a General Factor. *Journal of Adult Development*, *24*(1), 58–76. https://doi.org/10.1007/s10804-016-9247-4

Gahagan, K., Vaterlaus, J. M., & Frost, L. R. (2016). College student cyberbullying on social networking sites: Conceptualization, prevalence, and perceived bystander responsibility. *Computers in Human Behavior*, *55*, 1097–1105. https://doi.org/10.1016/j.chb.2015.11.019

Gray, K. L., Buyukozturk, B., & Hill, Z. G. (2017). Blurring the boundaries: Using Gamergate to examine "real" and symbolic violence against women in contemporary gaming culture. *Sociology Compass*, *11*(3), n/a–n/a. https://doi.org/10.1111/soc4.12458

Herring, S. C., Job-Sluder, K., Scheckler, R., & Barab, S. (2002). Searching for safety online: Managing "trolling" in a feminist forum.

Komac, G., & Cagiltay, K. (2019). An Overview of Trolling Behavior in Online Spaces and Gaming Context. *2019 1st International Informatics and Software Engineering Conference (UBMYK)*, 1–4. https://doi.org/10.1109/UBMYK48245.2019.8965625

Mihaylov, T., Mihaylova, T., Nakov, P., Màrquez, L., Georgiev, G. D., & Koychev, I. K. (2018). The dark side of news community forums: opinion manipulation trolls. *Internet Research*, *28*(5), 1292–1312. https://doi.org/10.1108/IntR-03-2017-0118

Nicol, S. (2012). Cyber-bullying and trolling. *Youth Studies Australia*, *31*(4), 3–4.

Thacker, S., & Griffiths, M. D. (2012). An Exploratory Study of Trolling in Online Video Gaming. *International Journal of Cyber Behavior, Psychology, and Learning*, *2*(4), 17–33. https://doi.org/10.4018/ijcbpl.2012100102

Thacker, S., & Griffiths, M. D. (2012). An Exploratory Study of Trolling in Online Video Gaming. *International Journal of Cyber Behavior, Psychology, and Learning*, *2*(4), 17–33. https://doi.org/10.4018/ijcbpl.2012100102

The Information Society, 18(5), 371-384.Marwick, A. E., & Lewis, R. (2017). Media manipulation and disinformation online. Data & Society, 15.

Zhao, Y., Chu, X., & Rong, K. (2023). Cyberbullying experience and bystander behavior in cyberbullying incidents: The serial mediating roles of perceived incident severity and empathy. *Computers in Human Behavior*, *138*, 107484. https://doi.org/10.1016/j.chb.2022.107484